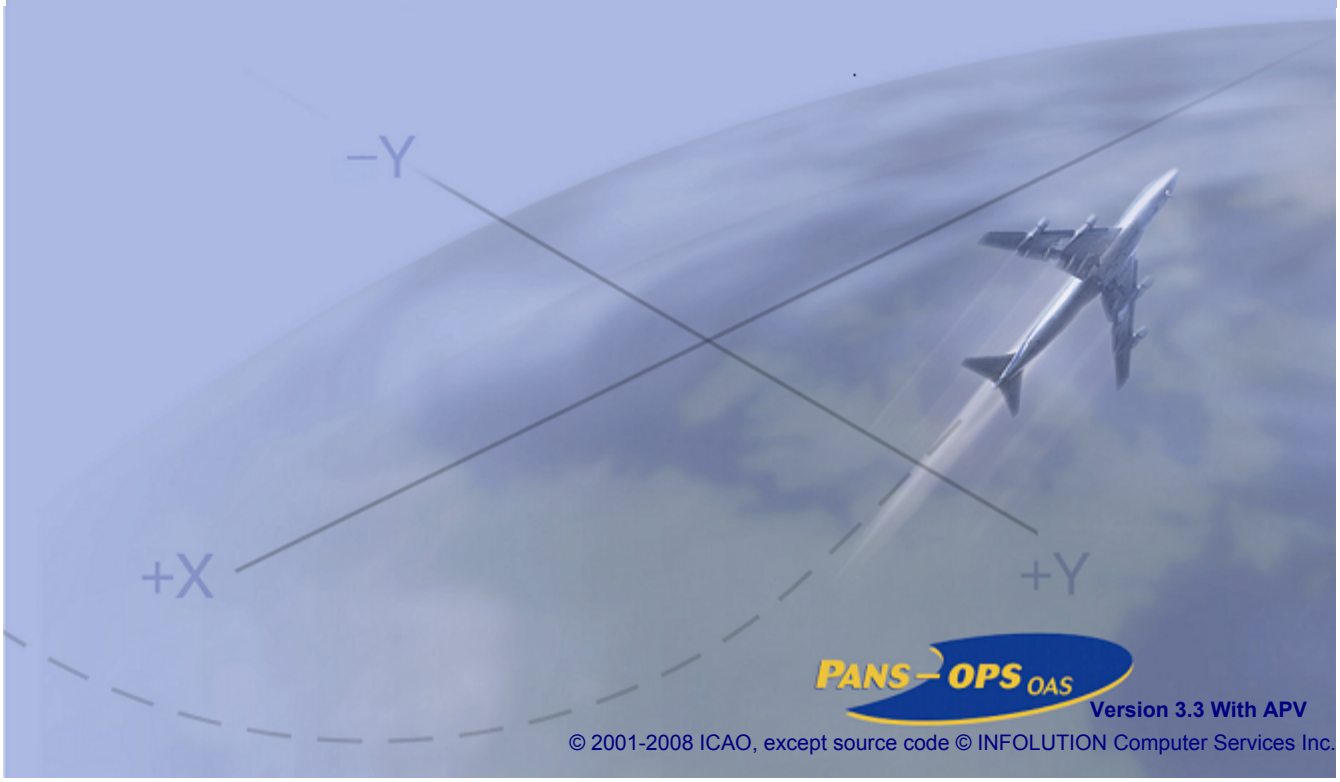
PANS-OPS OAS
User Guide



PANS-OPS OAS

Version 3.3 With APV

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PANS-OPS OAS

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PANS-OPS OAS

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Part I

GENERAL

1 GENERAL

This program provides the calculation of Obstacle Assessment Surfaces (OAS) parameters for specific ILS/MLS/GLS geometry, aircraft dimensions and missed approach climb gradient, and calculates the height (Z) of the OAS surface above a specific location (X,Y) for the selected system and aircraft parameters. The program prints the parameters and results for any specific set of conditions.

Editorial note. All references to PANS-OPS in this document are to PANS-OPS, Volume II, Fifth Edition.

1.1 Disclaimer

The user shall be aware that the Software should not be used without adequate knowledge of the contents of ICAO Document 8168-OPS/611, PROCEDURES FOR AIR NAVIGATION SERVICES, AIRCRAFT OPERATIONS, Volume II; and the user guide for the PANS-OPS OAS Software. The Software contains a tool developed to assist instrument procedure designers and is not intended to be a substitute for the approach designer's professional judgment.

Neither ICAO nor INFOLUTION Computer Services, Inc. shall be held liable for death or bodily injury, damage to property or any other direct, indirect or incidental damages or other loss sustained by third parties which may arise as a result of customer use of the Software, nor for damage inflicted with respect to any property of the customer or any other loss sustained by said customer. Neither INFOLUTION Computer Services, Inc., nor ICAO shall be responsible for the accuracy or validity of the data entered and/or generated by the Software. The customer shall be responsible for validation and verification of the instrument approach procedure designed using the Software, by means of flight checks or other verification mechanisms, in order to ensure the conformity of the instrument approach procedures in relation to the proper norms and standards.

1.2 System requirements

The **minimum** hardware and software requirements are:

Computer type:	Pentium™ 233 MHz
Operating system:	Microsoft Windows™ 95
Memory:	64 MB
Hard disk space:	20 MB
CD-ROM drive	
Mouse	

The screen resolution is designed for a minimum resolution of 800 x 600.

1.3 User guide conventions

To facilitate reading and comprehension of this user guide, note the following example of the conventions used:

Menu selection
Program | OAS means click on **Program** on the Taskbar and then click on **OAS** in the program sub-menu.

Main window
The main window contains the top title bar with the main menu name and the **Minimize**, **Maximize** and **Close** buttons.

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1.4 Program setup

Uninstall outdated version of PANS-OPS OAS using the Microsoft Windows "Add or Remove Programs" feature located in the Control Panel. Place the CD in the CD drive. If autorun is available, the install program will start automatically. Otherwise, browse to the CD and double click on **PANS-OPS_OAS_Setup.exe**. Then follow the on-screen instructions to install the program. It will add the program to your program group and place a shortcut on your desktop.

1.5 Program opening and closing

To start PANS-OPS OAS, click on the PANS-OPS OAS desktop icon or on the PANS-OPS OAS.exe program file in Windows Explorer, or **Start** in the Windows Taskbar and then select PANS-OPS OAS.exe as follows:

Programs | Infolution | PANS-OPS OAS | PANS-OPS OAS.exe

To close PANS-OPS OAS, click on the **close** button in the title bar of the active window, then click on the **close** button in the title bar of main program. *Note. The program cannot be closed until all sub-windows are closed.*



If the sub-window has been maximized, the buttons are positioned as follows:



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1.6 Limitations and warnings

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1.6.1 Undo and redo

The **undo** and **redo** functions (ALT-U or CTR-Z) have only one level. *Note. The **undo** symbol under the **Edit** menu does not change between **undo** and **redo**.*

1.6.2 Resolution

This program **does not detect the display resolution**, therefore fonts and module windows are smaller on high resolution displays. As a result, the screen may display several superimposed windows, the lower one having an empty white background. If the cursor is clicked on this white area while using any module, the blank background will be brought to the top level and obscure the module in use. This can be recognized by the blank screen and the presence of a small symbol (a solid rectangle partly hidden behind several dotted rectangles). Pressing the **ESC** key will recall the missing window.

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Part II

LANGUAGES

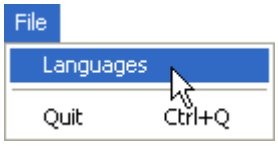
2 LANGUAGES

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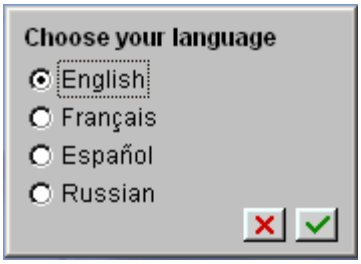
2.1 Languages menu

The languages menu allows the user to choose the language in which the PANS-OPS OAS software will be operated.

Click on **File | Languages** to obtain the **Languages menu** window.



The **Languages menu** allows the user to choose one of the four languages in which PANS-OPS is published.



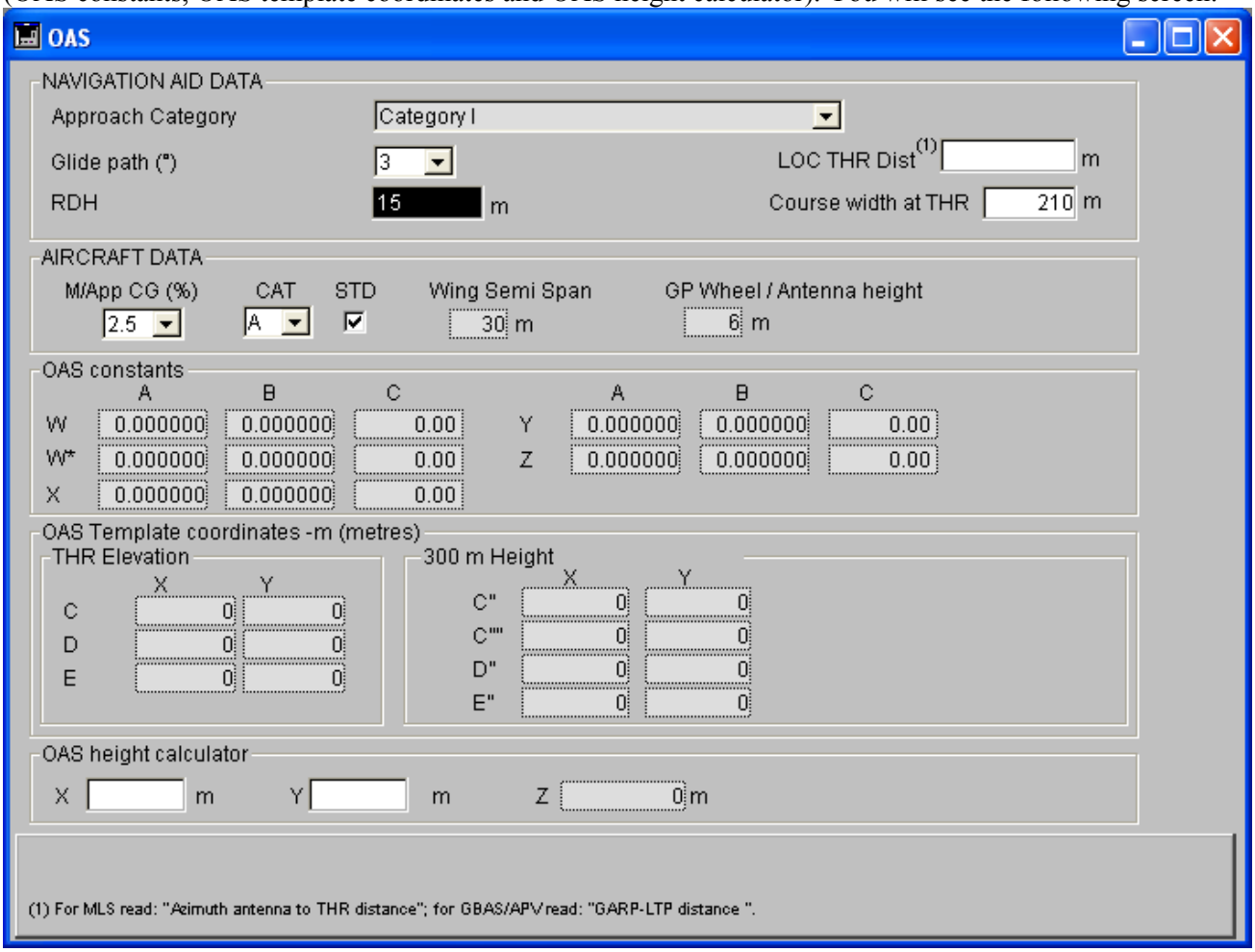
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Part III

OPENING THE PROGRAM

3 OPENING THE PROGRAM

Select **OAS** from the Program main menu bar to access the window for data entry and OAS output results (OAS constants, OAS template coordinates and OAS height calculator). You will see the following screen:



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Part IV

ENTRY OF DATA

4 ENTRY OF DATA

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4.1 General

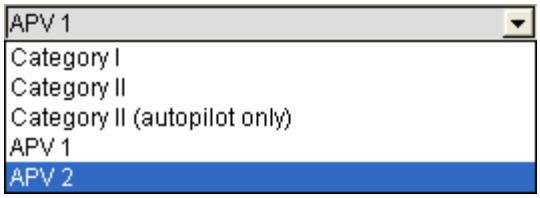
In order to enable any calculation, both the navigation aid and aircraft data need to be entered.

4.2 Navigation aid data

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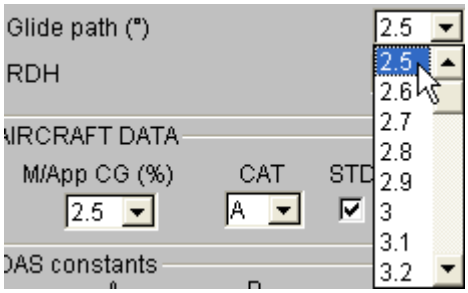
4.2.1 Approach category

The approach category of operation is selected from the drop-down list.



4.2.2 Glide path

The glide path angle is selected from the drop-down list. Values between 2.5 and 3.5 degrees may be selected in 0.1 degree increments.



4.2.3 Reference datum height (RDH)

The OAS constants and OAS height calculator are based on a reference datum height (RDH) of 15 m. An adjustment to the OAS constants is mandatory for an RDH less than 15 m and is optional for an RDH greater than 15 m. The RDH should be entered in the related box.



Warnings are given if the value entered is outside the range of 15 to 18 m in Category II, and 12 to 18 m in Category I (see Annex 10,). *Note. Observe the differences for microwave landing system (MLS) and GNSS landing system (GLS).*

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4.2.4 ILS localizer threshold distance (LOC THR Dist)

The ILS localizer threshold distance (LOC THR Dist) or equivalent MLS azimuth threshold and GBAS GARP-LTP distance is entered in the following box:



The localizer threshold distance should be between 2 000 m and 4 500 m. Extrapolation outside these limits is not permitted. If a localizer threshold distance outside this range is entered, the PANS-OPS OAS software will give the coefficients for 2 000 m or 4 500 m, as appropriate, which must be used.

4.2.5 Course width at THR

Course width at THR is a generic term denoting the sector width, approach azimuth deviation or GBAS course width for ILS, MLS or GLS applications, respectively.

A default value of 210 m appears when the program is started. If a value less than 210 m is entered, a warning is given and the Course width at THR defaults back to 210 m. If a value greater than 210 m is entered, a warning advises that the Collision Risk Model (CRM) method must be used and the constants and template coordinates all default to zero.



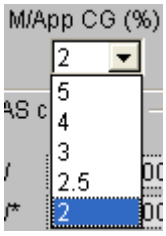
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4.3 Aircraft data

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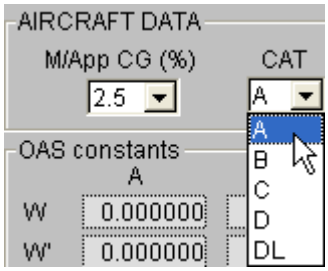
4.3.1 Missed approach climb gradient (M/APP CG)

The program adjusts all results for aircraft capable of missed approach climb gradients (M/APP CG) higher than the nominal 2.5 per cent. The desired missed approach climb gradient can be selected from the associated drop-down list.



4.3.2 Aircraft dimensions

First, select the aircraft category from the drop-down list.



The standard dimensions are then displayed in the adjacent boxes.



To obtain the constants and template coordinates or to calculate the height of the OAS for an aircraft differing from the standard dimensions given, de-select the **STD** box and enter the new dimensions.



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Part V

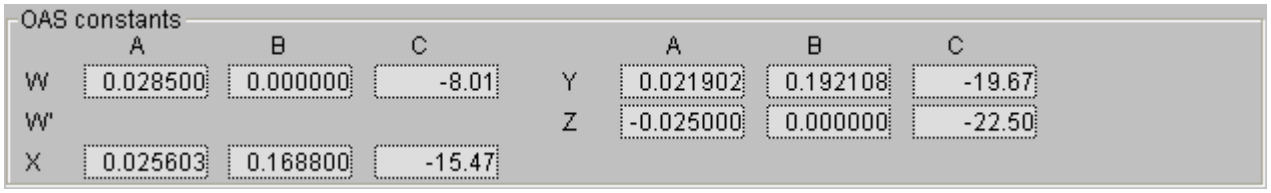
CALCULATION RESULTS

5 CALCULATION RESULTS

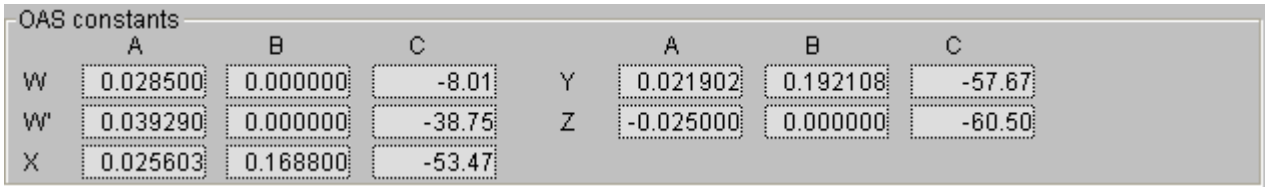
The program output consists of the OAS A, B and C constants for all surfaces, the template coordinates and the OAS height calculator. All of these incorporate the corrections for the settings made in the navigation and aircraft data boxes.

5.1 OAS constants

The geometry of the sloping surfaces is defined by four linear equations of the form z = Ax + By + C. In these equations x and y are position coordinates and z is the height of the surface at that position.



Note. For the Autopilot and APV, a further surface W'/W is added in the final approach.*



The constants shown already account for any adjustments made in the input parameters (RDH, aircraft size, etc.).

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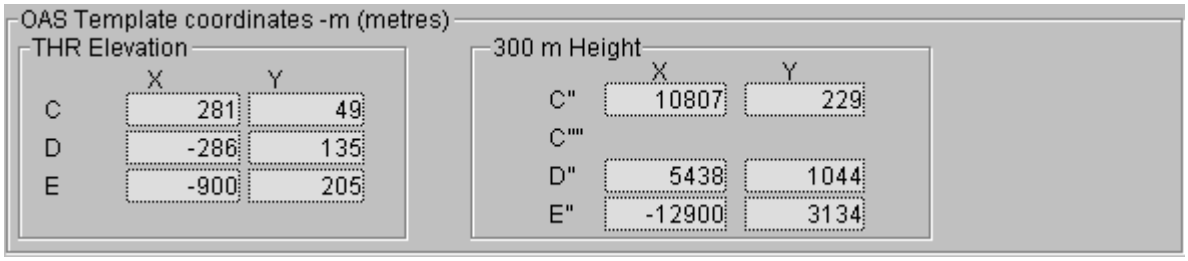
5.2 OAS template coordinates

Templates or plan views of the OAS contours to map scale are sometimes used to help identify obstacles for detail survey. The OAS template data includes:

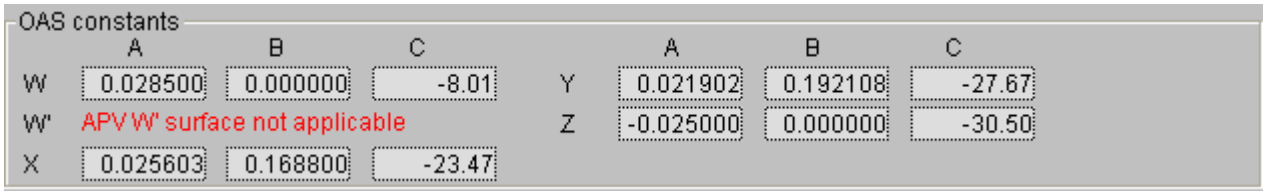
- The coordinates of the points of intersection of the sloping surfaces at threshold level. The intersection coordinates are labelled as C, D and E
- The coordinates at 300 m above threshold level for Category I
- The coordinates at 150 m for Category II

The intersection coordinates for 300/150 m (depending on the approach category selected) are labelled C", D" and E". The C"" coordinates relate to the intersection of the W and W'/W* surfaces for the APV and Autopilot Cat II cases.

The coordinates shown already account for any adjustments made in the input parameters (RDH, aircraft size, etc.).



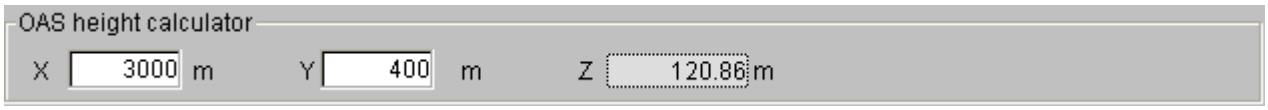
For APV II, in certain cases the W' surface is above the W surface at threshold level. In this case, the values of the W' surface constants and the C"" template coordinates are not displayed. A warning will appear.



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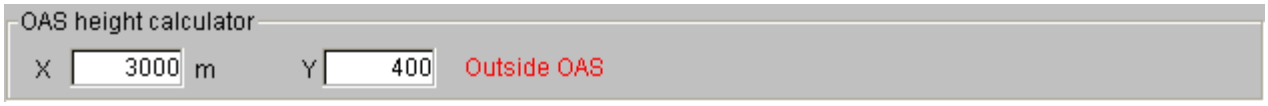
5.3 OAS height calculator

Enter the X and Y values for the obstacle being checked and see the resulting OAS height in the Z box.



Penetration of the obstacle can be easily checked by comparing the OAS height with the height of the obstacle.

If the location is outside the area covered by the OAS, a red warning "Outside OAS" appears.



5.4 Limitations

The user is responsible for calculating the longitudinal extent of the precision surfaces, both in the intermediate and initial approach when there is no FAF/descent fix, and in the missed approach when a turn is specified before the end of the precision segment. The calculator assumes:

- The W and X surfaces extend indefinitely in the positive direction
- The Y and Z surfaces terminate 300 m above threshold in Category I and APV, and 150 m above threshold in Category II

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Part VI

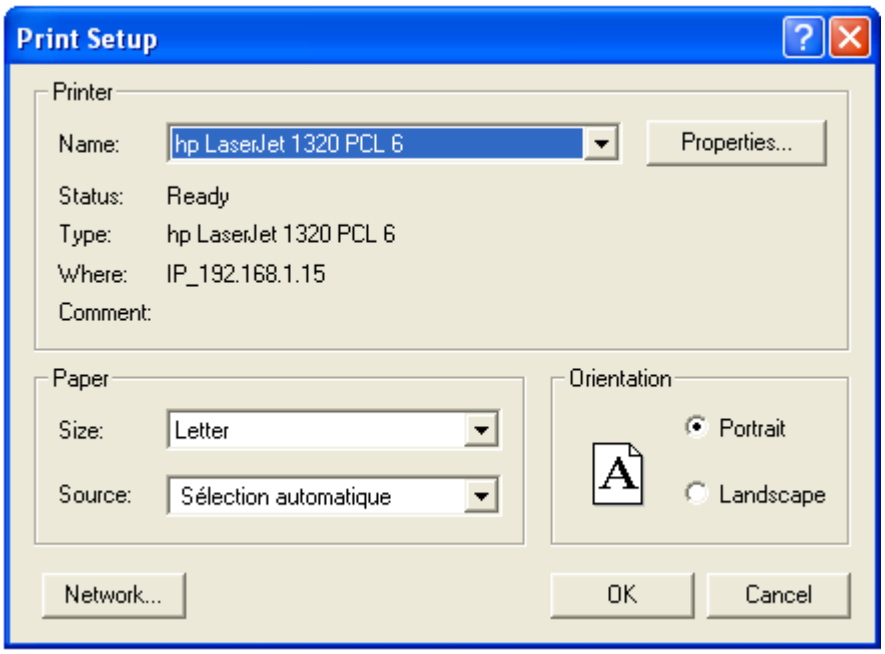
PRINTING

6 PRINTING

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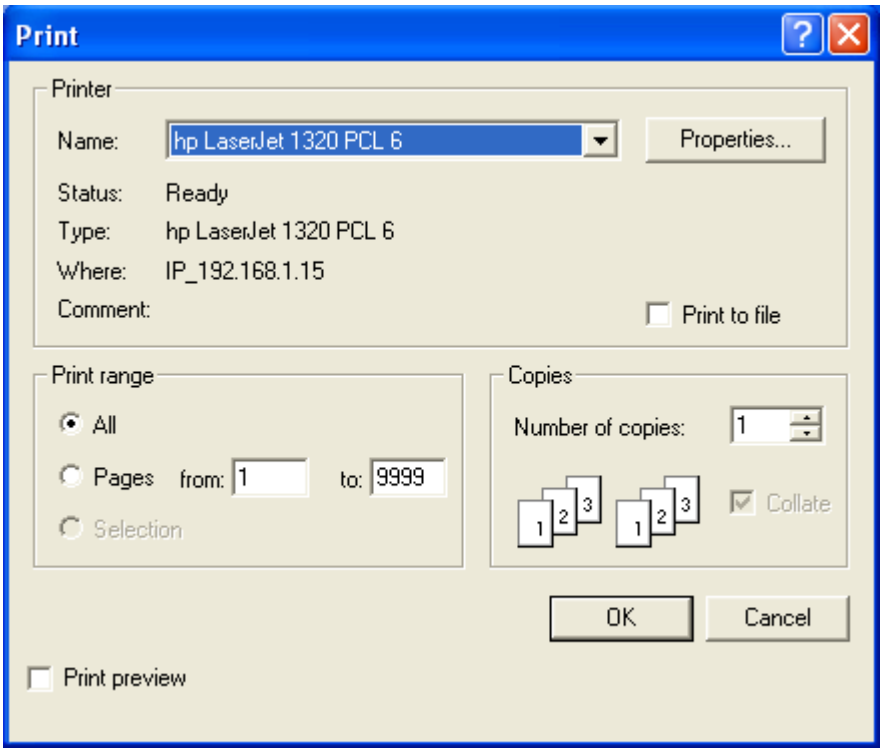
6.1 General

There are two basic options for printing calculation results and OAS tables. Both use the following two print control windows. The first is the **Print Setup** window.



Ensure **Landscape** is selected for printing calculation results and **Portrait** for printing OAS tables. Click **OK** to open the second print control window.

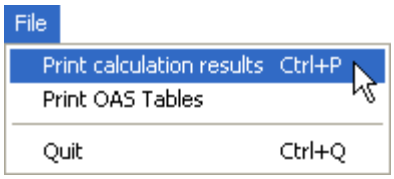
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WARNING: The Print range selector and the Print to file options DO NOT WORK in this version of the program. *Note. A **Print preview** box needs to be checked to enable printing of individual pages.*

6.2 Printing calculation results

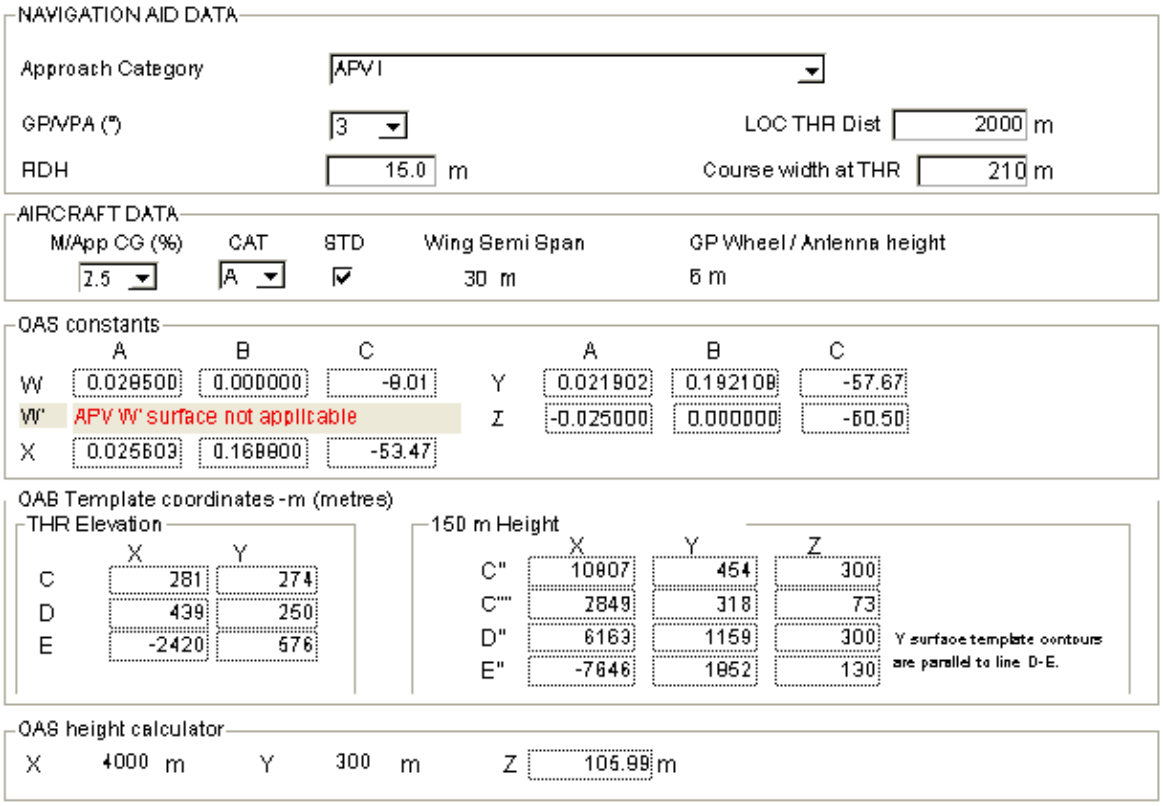
Select **Print calculation results** from the File menu bar.



The Print setup and Print windows then appear, allowing printing or preview of the calculation results.

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PANS-OPS OAS

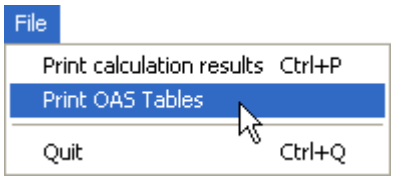


Note. Although the A and B constants are shown to only 6 decimal places, the OAS height calculator uses these constants at the full precision of the computer system. The template coordinates at threshold elevation are displayed to the nearest integer; while at 300 m above threshold elevation for Cat I, and 150 m for Cat II, the template coordinates are truncated.

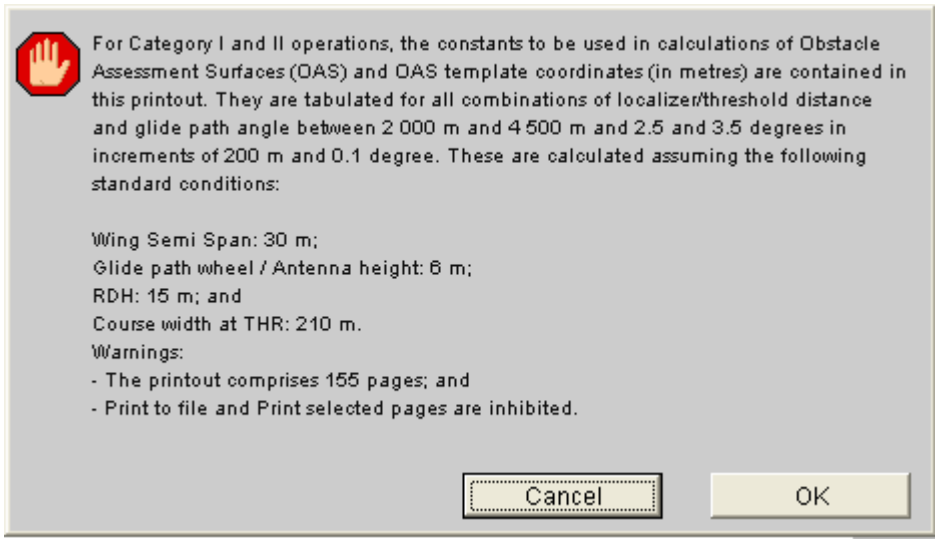
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6.3 Printing OAS tables

Select **Print OAS Tables** from the File menu bar.

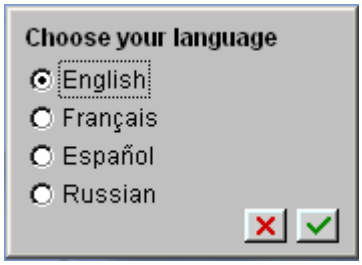


A warning appears:



Warning: Unless the **Print Preview box** is checked in the subsequent Print window, 155 pages will be sent to the printer. If only a few pages are needed, check the **Print Preview box** and click on the **Print button** in the Print Preview window to print individual pages.

Clicking **OK** moves to a box asking for the language required.

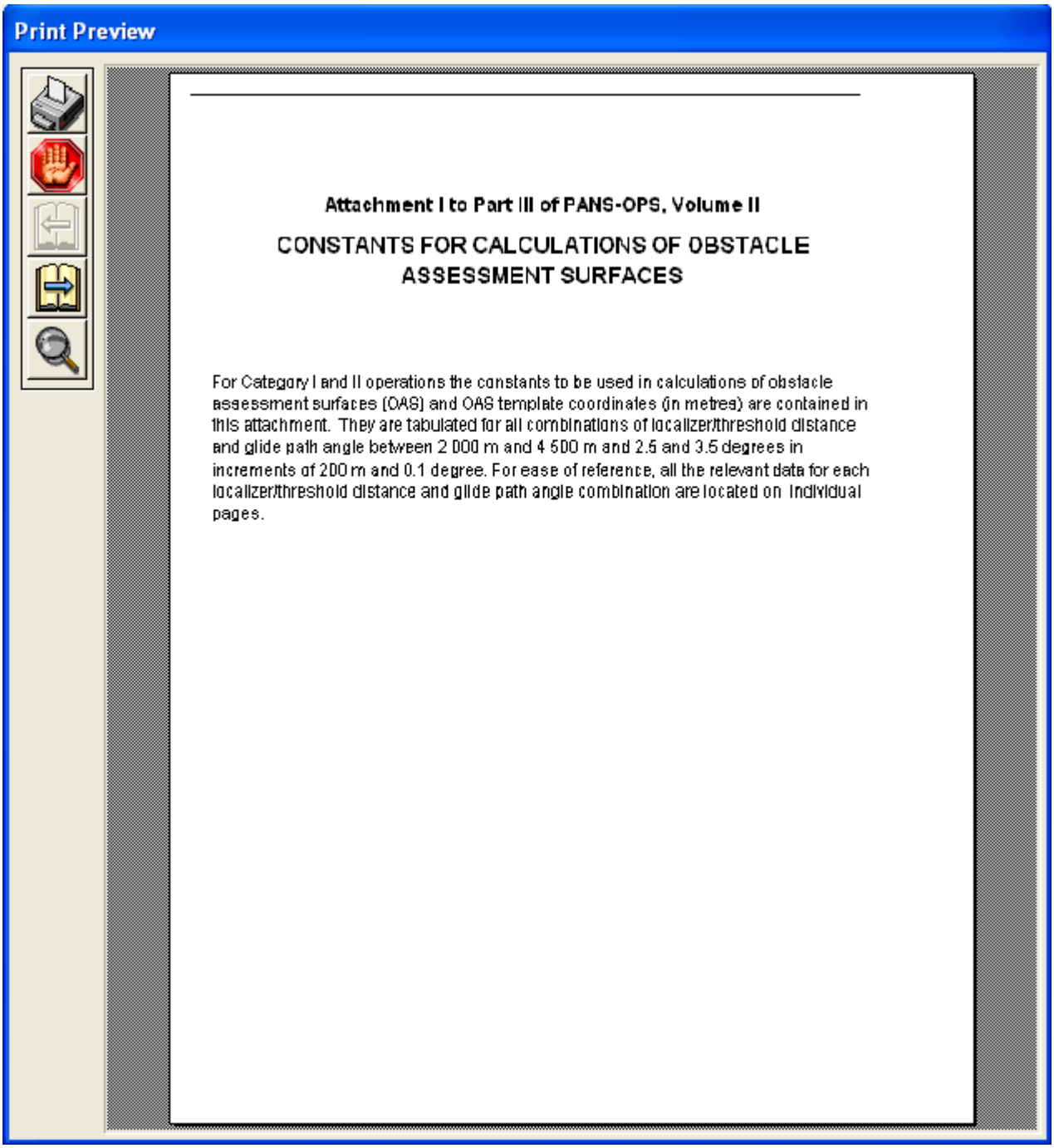


The program will print the complete tables formatted as the original Attachment I to PANS-OPS 4th Edition.

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6.3.1 OAS tables print preview

If the **Print Preview** box in the Print menu is checked, the Print Preview window appears.



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The buttons on the left of the Print Preview window are described as follows:

 Print current page

 Stop and return to main program

 Page back

 Page forward

 Magnify font

-END-

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